TOPS TECH MODUAL-4 Assessment

SECURITY AND MAINTENANCE

THEME: SMALL BUSINESS IT SETUP

SECTION A: Scenario-Based Concept Application

1. **Explain how the installation of Active Directory Domain Services (AD DS) facilitates centralized identity and access control for a small business IT setup.**

Answer:

- AD DS is a Microsoft service that acts as a central database to manage all users, computers, and resources on a network.
- Centralized Identity: All user accounts are stored on one Domain Controller. Any PC in the domain uses the same login — no need for separate accounts on each machine.
- Access Control: Administrators set permissions centrally — who can access which folder, printer, or application. Changes apply to all machines at once.
- Single Sign-On (SSO): Once a user logs into the domain, they can access all permitted resources without logging in again.
- Group Management: Users are grouped (e.g., HR Group, IT Group), and permissions are assigned to the group — saves time and reduces errors.
- Example: An employee logs in from any office PC and gets access only to their department's shared folder — all controlled from one server.

2. **Explain how the implementation of Group Policy Objects (GPOs) enforces robust password complexity and account lockout requirements across a domain.**

Answer:

- GPO is a set of rules pushed from the Domain Controller to all computers and users in the domain automatically.
- Password Complexity via GPO: Computer Configuration > Windows Settings > Security Settings > Account Policies > Password Policy. Settings enforced:
- Minimum password length: e.g., 8 characters
- Password complexity: must include uppercase, lowercase, numbers, and symbols
- Maximum password age: e.g., 90 days (forces regular password change)
- Account Lockout via GPO: Account Policies > Account Lockout Policy. Settings:
- Lockout threshold: e.g., 5 failed login attempts
- Lockout duration: e.g., 30 minutes
- Reset counter after: e.g., 15 minutes
- How it works: Once the GPO is linked to an OU or domain, all machines automatically receive and apply these rules without any manual setup on each PC.

3. **Explain the interaction between NTFS permissions and Shared Folder permissions when managing multi-departmental access to sensitive company data.**

<u>Answer:</u>

- <u>NTFS Permissions:</u> Applied at the file/folder level on disk. Controls Read, Write, Modify, or Full Control. Apply both locally and over the network.
- <u>Shared Folder Permissions:</u> Applied only when accessing a folder over the network. Options are Read, Change, or Full Control.

- The Key Rule - Most Restrictive Wins: When both NTFS and Share permissions apply, the effective permission is whichever is more restrictive.
- Example:
- Share Permission for HR Group = Full Control
- NTFS Permission for HR Group = Read Only
- Result = User gets Read Only (NTFS is stricter)
- Best Practice: Set Share Permission to Full Control for Authenticated Users, then control access tightly using NTFS permissions per department group. NTFS becomes the real gatekeeper.

4. **Explain the DHCP DORA process (Discover, Offer, Request, Acknowledge) and how it eliminates manual IP address conflicts for network users.**

<u>Answer:</u>

- DORA is the 4-step automatic process a DHCP server uses to assign IP addresses to devices:
- D - Discover: Client PC broadcasts: 'Is there a DHCP server? I need an IP address.' Sent to 255.255.255.255 (broadcast to all).
- O - Offer: DHCP server replies: 'I have IP 192.168.1.50 available. Here are the details.' (Offers IP + subnet mask + gateway + DNS)
- R - Request: Client replies: 'Yes, I accept that IP from you.' (Broadcast so other DHCP servers know this offer was accepted)
- A - Acknowledge: DHCP server confirms: 'The IP is yours for the lease duration.' The client now uses that IP.
- Eliminates Conflicts: DHCP tracks all assigned IPs in a lease database. It never gives the same IP to two devices. Manual IP assignment often causes duplicate IPs — DHCP automation prevents this entirely.

5. **Explain the role of DNS Name Resolution in ensuring internal users can consistently access the company's hosted web resources.**

Answer:

- DNS translates human-readable names (like fileserver.company.local) into IP addresses that computers use to communicate.
- Without DNS: Users would have to remember IP addresses like 192.168.1.10 to reach the file server. Any IP change would break access for everyone.
- Internal DNS Role:
- The Windows Server DNS holds A records (name-to-IP mappings) for internal servers.
- When a user types \fileserver or http://intranet, DNS resolves it to the correct IP automatically.
- If the server IP changes, the admin updates only the DNS record — users keep using the same name with no interruption.
- Consistency benefit: DNS names remain stable even when server IPs change. Users always reach the correct resource without reconfiguration on their PCs.
- Example: User types http://intranet.technova.local — DNS resolves to 192.168.1.20 — browser connects. No need for users to know the IP.

6. **Explain the technical configuration of a Windows Server Backup Schedule to ensure automated, regular data protection without manual intervention.**

Answer:

- Windows Server Backup is a built-in tool (wbadmin) that automates regular backups of data and system state.
- Step 1: Install feature via Server Manager > Add Roles and Features > Features > Windows Server Backup.
- Step 2: Open Windows Server Backup from Administrative Tools.
- Step 3: Click Backup Schedule in the right panel.
- Step 4: Choose backup type: Full Server (recommended) or Custom (specific volumes/folders).
- Step 5: Set backup time — e.g., Daily at 11:00 PM (off-peak hours to avoid slowdown).
- Step 6: Choose destination — a dedicated backup drive or network share. Never back up to the same disk.
- Step 7: Click Finish — the schedule is saved and runs automatically every day.
- CLI alternative:

```
wbadmin enable backup -addtarget:{BackupDrive} -schedule:23:00 -allCritical -quiet
```

- This runs a full critical-volume backup every night at 11 PM automatically with no manual action needed.

SECTION B: Practical Application

Task 1 (Core Services): Install Windows Server and promote it to a Domain Controller with AD DS and DNS.

Answer:

- Step 1: Install Windows Server 2022 in VirtualBox/VMware. Set a static IP (e.g., 192.168.1.1) and set DNS to itself (127.0.0.1).
- Step 2: Open Server Manager > Add Roles and Features.
- Step 3: Select Active Directory Domain Services + DNS Server > Install.
- Step 4: After installation, click the flag notification > Promote this server to a domain controller.
- Step 5: Choose Add a new forest > enter domain name (e.g., technova.local).
- Step 6: Set DSRM password > proceed through wizard > click Install.
- Step 7: Server reboots automatically. After reboot, it is a Domain Controller with AD DS and DNS active.

Task 2 (Directory Structure): Create 3 Organizational Units (OUs), 5 User Accounts, and 2 Security Groups via Active Directory Administrative Center.

Answer:

- Open: Server Manager > Tools > Active Directory Administrative Center (ADAC).
- Create 3 OUs: Right-click domain name > New > Organizational Unit. Create: HR_OU, IT_OU, Finance_OU.
- Create 5 User Accounts: Right-click an OU > New > User. Enter first name, last name, logon name, and password. Example users: john.hr (HR_OU), alice.it (IT_OU), bob.finance (Finance_OU), jane.hr (HR_OU), mike.it (IT_OU).
- Create 2 Security Groups: Right-click OU > New > Group. Set Group type = Security, Scope = Global. Create: HR_Group and IT_Group. Add relevant users to each group.

Task 3 (Policy Enforcement): Configure a GPO to enforce complex passwords and restrict Control Panel access for specific OUs.

Answer:

- Open: Server Manager > Tools > Group Policy Management.
- Right-click HR_OU > Create a GPO in this domain and link it here. Name it: Security_Policy_GPO.
- Right-click the GPO > Edit.
- Password Complexity: Computer Configuration > Policies > Windows Settings > Security Settings > Account Policies > Password Policy. Set: Password must meet complexity = Enabled. Minimum length = 8.
- Restrict Control Panel: User Configuration > Policies > Administrative Templates > Control Panel. Set: Prohibit access to Control Panel and PC settings = Enabled.
- Close editor. GPO applies to all users in HR_OU at next login or after gpupdate /force.

Task 4 (Resource Management): Create a shared folder on an NTFS volume with granular NTFS and Sharing permissions for defined groups.

Answer:

- Step 1: On the server, create a folder: C:\CompanyData\ProjectFiles.
- Step 2 - NTFS Permissions: Right-click folder > Properties > Security > Edit. Add HR_Group: Read & Execute. Add IT_Group: Modify. Remove 'Everyone'.
- Step 3 - Share the folder: Properties > Sharing > Advanced Sharing > Check 'Share this folder'. Name it ProjectFiles.
- Step 4 - Share Permissions: Click Permissions > Add HR_Group and IT_Group with Change permission. Remove Everyone.

- Step 5: Users access via \\ServerName\ProjectFiles. Effective access = the most restrictive of NTFS and Share permissions.

Task 5 (Network Automation): Authorize a DHCP scope to automate IP leasing and verify connectivity using ipconfig /renew on a client.

Answer:

- Step 1: Open Server Manager > Tools > DHCP.
- Step 2: Right-click IPv4 > New Scope. Enter: Name, Start IP (192.168.1.100), End IP (192.168.1.200), Subnet Mask (255.255.255.0).
- Step 3: Set exclusions (e.g., 192.168.1.1 to 192.168.1.10 for static devices).
- Step 4: Set lease duration. Add gateway (192.168.1.1) and DNS (192.168.1.1).
- Step 5: Activate scope. Right-click the DHCP server name > Authorize.
- Step 6: On client PC, open Command Prompt and run:

```
ipconfig /release
ipconfig /renew
```

- Step 7: Run ipconfig to verify the client received an IP from the scope range. Ping the gateway to confirm connectivity.

SECTION C: Integrated Case Study

Scenario:

'TechNova,' a boutique marketing firm, recently updated its Group Policy Objects (GPOs) to enhance security. Immediately following the update, several employees reported that while they can log into their workstations, they are met with 'Access Denied' errors when attempting to reach the centralized Project Files share. The administrator suspects a conflict between the new security templates and existing NTFS permissions, or a failure in GPO replication across the network.

C-Q1. Analysis: Identify three specific GPO settings or NTFS conflicts that could cause users to log in successfully but lose access to shared drives.

Answer: Three likely causes:

- 1. GPO Restricted Groups Setting: The new GPO may have overwritten group memberships. A 'Restricted Groups' policy could have removed users from HR_Group (which had NTFS access) — so they log in fine but lose share access because they are no longer in the permitted group.
- 2. NTFS Permission Overwrite via Security Template: A new GPO security template may have reset NTFS permissions on the ProjectFiles folder — replacing custom permissions with defaults (Administrators only), locking out regular users.
- 3. GPO Replication Failure: If the GPO was updated on one Domain Controller but not fully replicated to others, some users receive the new broken policy while others do not — causing inconsistent 'Access Denied' errors across the network.

C-Q2. Troubleshooting: Detail the step-by-step process of using the Resultant Set of Policy (RSOP) or the gpresult tool to diagnose this issue.

Answer:

- Method 1 - gpresult (CLI):
- Step 1: Log in to an affected user's PC. Open Command Prompt.
- Step 2: Run:

```
gpresult /R
```

- Shows Applied GPOs, Denied GPOs, and which OU/domain the policy came from.

- Step 3: For a full HTML report:

```
gpresult /H C:\GPOReport.html /F
```

- Open in browser. Review the Computer Configuration and User Configuration sections for any security or access policies blocking the share.
- Method 2 - RSOP (GUI):
- Step 1: Press Win + R > type rsop.msc > Enter.
- Step 2: RSOP loads automatically for the current user/computer — showing all applied policy settings.
- Step 3: Navigate to: Computer Configuration > Windows Settings > Security Settings > File System — check if any GPO is redefining NTFS permissions on the shared folder path.
- Step 4: Navigate to: User Configuration > Administrative Templates — check for policies restricting network access or mapped drives.
- Step 5: Compare the RSOP output against the intended policy to find the setting causing Access Denied.

C-Q3. Tooling: List the specific CLI commands and GUI consoles you would use to verify if the file server is correctly seeing the user's updated security tokens.

Answer:

- CLI Commands:
- 1. Check the user's group membership on the Domain Controller:

```
net user john.hr /domain
```

- Shows all groups the user belongs to. Verify HR_Group is listed.
- 2. Check current groups in the user's active login token:

```
whoami /groups
```

- Lists all security groups in the current session token. If HR_Group is missing, the token is stale — the user must log off and log back on.
- 3. Force GPO refresh immediately:

```
gpupdate /force
```

- 4. Check NTFS permissions on the folder (on file server):

```
icacls C:\CompanyData\ProjectFiles
```

- Lists all NTFS permissions. Verify HR_Group and IT_Group have correct access.
- GUI Consoles:
- Active Directory Users and Computers (dsa.msc) — verify group membership
- Group Policy Management Console (gpmc.msc) — check GPO links and WMI filters
- Event Viewer > Security Log — check Event ID 5145 for Access Denied audit entries

C-Q4. Resolution: Propose a final fix to restore access and a preventive strategy, such as A-G-DL-P, to ensure future policy changes do not break permissions.

Answer:

- Immediate Fix:
- Open ADUC (dsa.msc) — verify HR_Group and IT_Group still have correct users assigned.
- On the file server, reapply correct NTFS permissions:

```
icacls C:\CompanyData\ProjectFiles /grant HR_Group:(RX)
icacls C:\CompanyData\ProjectFiles /grant IT_Group:(M)
```

- Force GPO refresh on all clients: gpupdate /force

- Have affected users log off and log back on to get a fresh security token.
- Preventive Strategy - A-G-DL-P Model:
- A = User Accounts are placed into...
- G = Global Groups (e.g., HR_Global_Group — contains all HR users)
- DL = Domain Local Groups (e.g., ProjectFiles_DL — this group is granted NTFS permission on the folder)
- P = Permissions are assigned only to the Domain Local Group
- Why this helps: NTFS permissions are set only on the DL group. User/group changes happen at the Global group level. Future GPO changes never touch the DL-P layer, preventing accidental permission loss.

C-Q5. Validation: Explain how to use the net share and icacls commands to confirm that the folder permissions match the intended business logic.

Answer:

- Step 1 - Verify the share exists using net share:

```
net share
```

- Lists all shared folders on the server. Confirm ProjectFiles share is listed.

```
net share ProjectFiles
```

- Shows share path and share-level permissions. Verify HR_Group and IT_Group have Change permission — not Everyone.
- Step 2 - Verify NTFS permissions using icacls:

```
icacls C:\CompanyData\ProjectFiles
```

- Expected output:

```
HR_Group:(RX)           Read & Execute
IT_Group:(M)            Modify
BUILTIN\Administrators:(F) Full Control
```

- Step 3 - Validate effective permissions for a specific user:
- Right-click folder > Properties > Security > Advanced > Effective Access tab. Enter username — it shows exactly what that user can do.
- Conclusion: If net share shows correct share permissions AND icacls shows correct NTFS permissions AND effective access matches business requirements — the configuration is correct, and the issue is fully resolved.